

# Where Does Scoring Bias Come From?

A Base-vs-Instruct Study of Three Scoring Biases in Small Open-Weight LLM Judges

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## Abstract

LLMs are widely used as automated judges, yet they show systematic *scoring biases*: their ratings shift when superficial features of the scoring task are perturbed. Li et al. (2026) documented three such biases—rubric order, score identifier, and reference-answer bias—but left open whether they originate in pre-training or are introduced during instruction tuning. We take a first empirical step on this question using a controlled base-vs-instruct comparison across seven open-weight model families ( $\leq 7$ B parameters) run on a free GPU. Treating each family as one paired observation ( $n = 7$ ), we find that instruction tuning *reduces* bias for all three probes. The reduction is large and robust for score-ID bias ( $d_z = -1.08$ , 6/7 families, bootstrap 95% CI  $[-1.54, -0.33]$ ) and reference-answer bias ( $d_z = -1.18$ , 6/7,  $[-1.27, -0.31]$ ), and weak and outlier-driven for rubric-order bias. We do *not* reproduce a “content bias increases with tuning” pattern in this regime. Our result is consistent with scoring bias being modulated at the instruction-tuning stage rather than fixed at pre-training, at least for small models, and suggests instruction tuning is net-beneficial for judge robustness here. We release a single-command pipeline that regenerates every number and figure from one data file.

## 1 Introduction

When one language model scores another’s output, we would like the score to reflect response quality—not the wording of the rubric or the presence of an example. Yet LLM judges are systematically sensitive to such surface features. Li et al. (2026) introduced and quantified three *scoring biases*: **rubric order** (does reversing the “1=worst...5=best” scale change scores?), **score ID** (do numeric vs. letter vs. descriptive labels change scores?), and **reference answer** (does showing a good or poor exemplar shift subsequent scores?). They observed these biases across five frontier models but explicitly noted that “the underlying causes of these scoring biases remain to be validated.”

This paper takes one concrete step toward a cause. We ask a single, tractable question: *are these scoring biases already present in pre-trained (base) models, or are they shaped by instruction tuning (SFT/RLHF)?* The answer matters for mitigation—if bias is baked in at pre-training, mitigation must target data and architecture; if it is shaped during alignment, mitigation can target the tuning stage. Following the base-vs-instruct methodology validated by Pan et al. (2026) and Thakur et al. (2024) for other judge behaviours, we compare the two variants of the same model family under identical probes.

**Contributions.**

1. A controlled base-vs-instruct comparison of all three Li et al. (2026) scoring biases across seven open-weight families ( $\leq 7B$ ), reproducible at zero cost.
2. The finding that instruction tuning *reduces* scoring bias in this regime—robustly for score-ID and reference-answer bias—rather than introducing it.
3. A fully reproducible pipeline: every table and figure is regenerated by one script from one data file, with no manual steps.

**Scope.** This is a deliberately narrow, exploratory study. Our inferential unit is the model *family* ( $n = 7$ ), all models are  $\leq 7B$ , and only English prompts are used. We state these limits plainly (§6) and frame the result as directional evidence, not a definitive causal claim.

## 2 Related Work

**Bias in LLM judges.** Wang et al. (2024) showed GPT-4 flips its preferred answer 46.3% of the time when response order is swapped; Zheng et al. (2023) systematised LLM-as-a-judge evaluation and noted length and position biases; Ye et al. (2024) catalogued twelve bias types; and Chen et al. (2024) found humans and LLMs share susceptibility to irrelevant factors. Shi et al. (2025) give a systematic account of position bias, and Gu et al. (2026) survey the area. Park et al. (2024) mitigate bias by tuning on debiased data.

**Scoring bias and its causes.** Li et al. (2026) introduced the three scoring biases we study and called for root-cause analysis. Thakur et al. (2024) observed that base and instruct Llama-2 differ as judges, and Pan et al. (2026) used a base-vs-instruct design to show that instruction tuning *introduces* a user-deference bias absent in base models. Our work applies the same design to Li et al. (2026)’s scoring biases, and finds the opposite sign for these particular biases: tuning reduces, rather than introduces, them.

## 3 Method

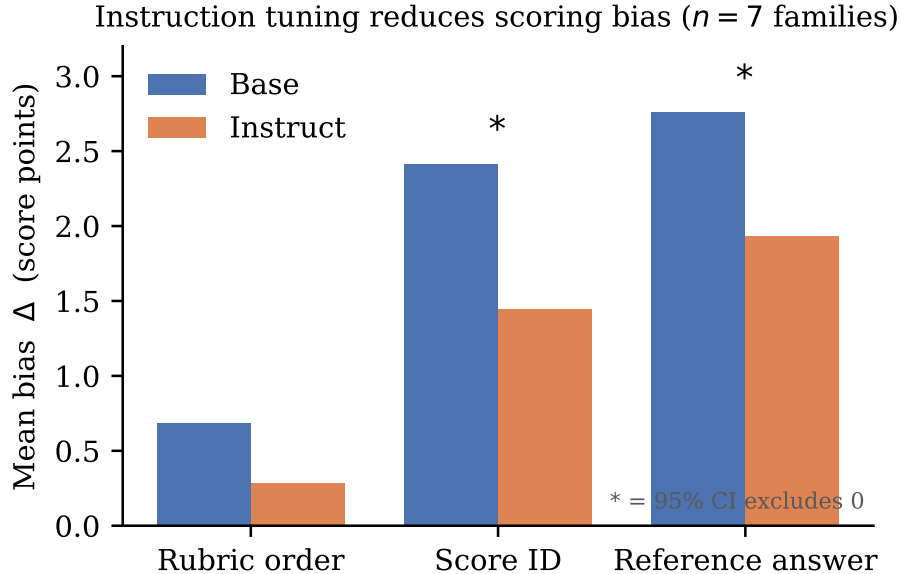
**Models.** We use seven open-weight families for which both a pre-trained (base) and an instruction-tuned checkpoint are publicly available: Qwen2.5-0.5B/1.5B/7B (Qwen Team, Alibaba Group, 2024), Llama-3.2-1B/3B (Meta AI, 2024), Gemma-2-2B (Gemma Team, Google DeepMind, 2024), and StableLM-2-1.6B (Bellagente et al., 2024). This spans 0.5–7B parameters and two alignment recipes (RLHF for six families, SFT for StableLM-2). All models are run with greedy decoding (temperature 0) on a free Kaggle T4 GPU.

**Probes.** We use the three perturbation probes of Li et al. (2026). **Rubric order:** normal (1=worst...5=best) vs. reversed. **Score ID:** numeric (1–5) vs. letter (A–E) vs. descriptive (Poor–Excellent). **Reference answer:** no exemplar vs. a good exemplar vs. a poor exemplar shown before scoring. Each model scores the same set of instruction–response items under each variant.

**Bias metric.** Following Li et al. (2026), we summarise each model $\times$ probe by  $\Delta$ , the maximum spread in mean score across that probe’s variants:  $\Delta = \max_v \bar{s}_v - \min_v \bar{s}_v$ . Larger  $\Delta$  means the model’s scores move more when only the surface feature changes, i.e. more bias.  $\Delta$  is bounded in  $[0, 4]$  on a 1–5 scale.

**Table 1:** Effect of instruction tuning on each scoring bias, across  $n = 7$  families.  $\Delta$  is the max inter-variant score spread (lower = less biased). **Bold** change = bootstrap 95% CI excludes zero.

| Probe            | Base<br>mean $\Delta$ | Instruct<br>mean $\Delta$ | Change<br>(%)       | $d_z$ | 95% CI<br>(bootstrap) |
|------------------|-----------------------|---------------------------|---------------------|-------|-----------------------|
| Rubric order     | 0.69                  | 0.29                      | -0.40 (-58%)        | -0.38 | [-1.21, +0.13]        |
| Score ID         | 2.41                  | 1.44                      | <b>-0.97 (-40%)</b> | -1.08 | [-1.54, -0.33]        |
| Reference answer | 2.76                  | 1.93                      | <b>-0.83 (-30%)</b> | -1.18 | [-1.27, -0.31]        |



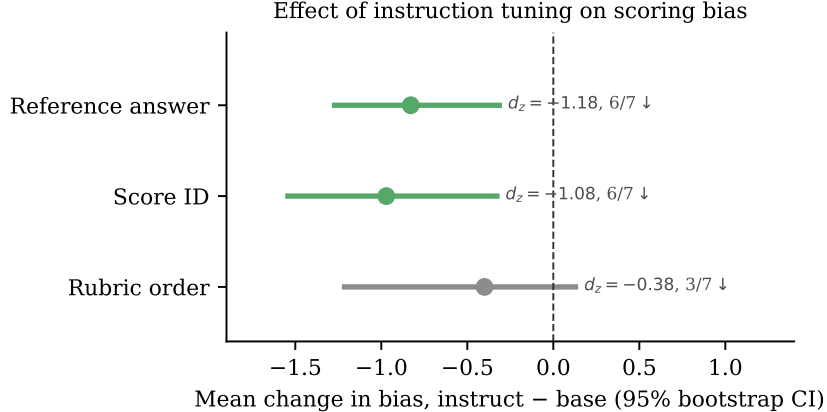
**Figure 1:** Mean bias before and after instruction tuning. Score-ID and reference-answer bias fall substantially; rubric-order bias is small in both. \* marks a bootstrap 95% CI on the change that excludes zero.

**Unit of analysis and inference.** The data of record retains, for each model and probe variant, the *mean* score across items—not the individual item scores.<sup>1</sup> We therefore analyse at the level of the model *family*: each family contributes one base  $\Delta$  and one instruct  $\Delta$  per probe, giving  $n = 7$  paired observations. For each probe we report the paired mean change (instruct – base), the standardized paired effect size  $d_z$ , a percentile bootstrap 95% CI of the mean change ( $10^4$  resamples, seed 42), the two-sided exact Wilcoxon signed-rank test, and the number of families in which bias decreased.

## 4 Results

**Instruction tuning reduces all three biases.** Table 1 and Figure 1 show the headline result. Mean bias falls from base to instruct for every probe. For **score-ID** bias the drop is  $-0.97$  points ( $-40\%$ ;  $d_z = -1.08$ ; 6/7 families; bootstrap CI  $[-1.54, -0.33]$ ), and for **reference-answer** bias it is  $-0.83$  points ( $-30\%$ ;  $d_z = -1.18$ ; 6/7;  $[-1.27, -0.31]$ ). Both CIs exclude zero and the exact Wilcoxon test attains its minimum two-sided  $p = 0.047$  at  $n = 7$ . **Rubric-order** bias also decreases

<sup>1</sup>The scores were transcribed from the run logs of the original experiment; per-item scores were not preserved. Consequently item-level quantities used elsewhere in the literature—flip rate, per-item discrimination, and domain breakdowns—are not computable here, and we do not report them.



**Figure 2:** Paired change in bias (instruct – base) with 95% bootstrap CIs. Green (CI excludes zero) marks robust reductions; grey marks the inconclusive rubric-order effect.

**Table 2:** Per-family bias  $\Delta$  before/after instruction tuning. B = parameters (billions); Train = alignment recipe of the instruct checkpoint.

| Family          | B   | Train | Rubric order |      | Score ID |      | Reference answer |      |
|-----------------|-----|-------|--------------|------|----------|------|------------------|------|
|                 |     |       | base         | inst | base     | inst | base             | inst |
| Qwen2.5-0.5B    | 0.5 | RLHF  | 0.1          | 0.2  | 1.3      | 0.4  | 2.6              | 1.2  |
| Qwen2.5-1.5B    | 1.5 | RLHF  | 0.1          | 0.4  | 3.1      | 1.8  | 2.6              | 1.8  |
| Llama-3.2-1B    | 1   | RLHF  | 0.0          | 0.2  | 1.3      | 2.0  | 2.2              | 2.6  |
| Llama-3.2-3B    | 3   | RLHF  | 3.5          | 0.8  | 3.7      | 2.4  | 3.1              | 2.5  |
| Gemma-2-2B      | 2   | RLHF  | 0.1          | 0.1  | 2.1      | 0.6  | 2.6              | 1.1  |
| StableLM-2-1.6B | 1.6 | SFT   | 0.4          | 0.3  | 2.9      | 2.5  | 3.8              | 3.4  |
| Qwen2.5-7B      | 7   | RLHF  | 0.6          | 0.0  | 2.5      | 0.4  | 2.4              | 0.9  |

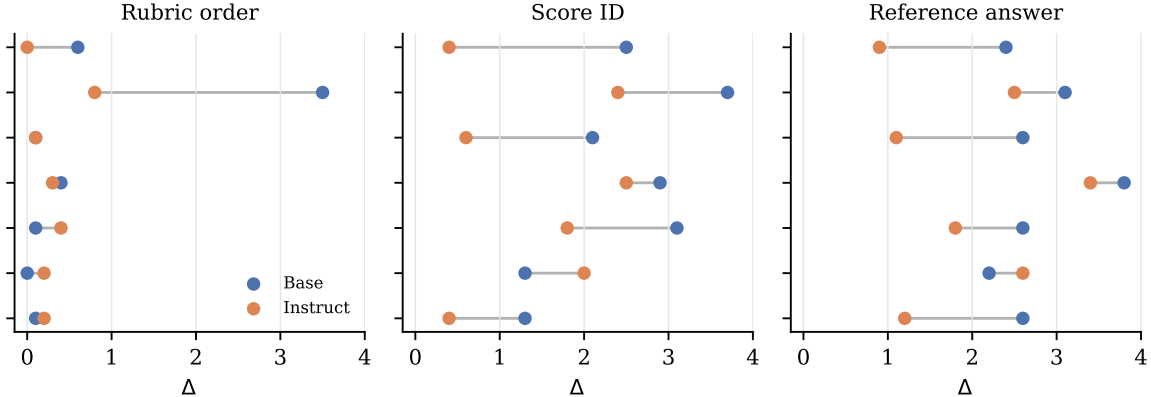
on average ( $-0.40$ ), but the effect is small, not robust (CI  $[-1.21, 0.13]$  includes zero, 3/7 families), and is dominated by a single family (Llama-3.2-3B, base  $\Delta = 3.5$  vs. a median base  $\Delta$  of only 0.1): most small base models simply are not sensitive to scale reversal.

**The reduction is consistent across families.** Figure 3 shows every family individually. For score-ID and reference-answer bias the instruct checkpoint is less biased than its base in six of seven families—the lone exception for both is Llama-3.2-1B, the smallest Llama model. The consistency of direction, more than the borderline  $p$ -value, is what supports the finding: with  $n = 7$  and three probes, we do not claim significance survives a multiple-comparison correction, and we treat the effect sizes and per-family agreement as the primary evidence.

## 5 Discussion

**Interpretation.** Base models can be strikingly biased: several assign sharply different scores under a mere relabelling (score ID) or after seeing an unrelated exemplar (reference answer). Instruction tuning consistently damps both. The most parsimonious reading is that these two biases are not immutable properties of the pre-trained network but are substantially *shaped by the instruction-tuning stage*—the same stage at which the model learns to follow the scoring instruction in the first place. This directly addresses the open question of Li et al. (2026) for small models: the cause is, at least in part, downstream of pre-training, so mitigation targeting the tuning stage

### Per-family bias: base $\rightarrow$ instruct



**Figure 3:** Per-family bias, base (●, blue)  $\rightarrow$  instruct (●, orange), sorted by size. Leftward movement = bias reduced by instruction tuning. Score-ID and reference-answer panels move left in 6/7 families each.

is plausible.

**Relation to prior sign of the effect.** Pan et al. (2026) found instruction tuning *adds* a bias (user deference). We find it *removes* scoring biases. These are compatible: instruction tuning teaches a model to parse and follow the scoring task more faithfully—helping with format-driven biases like score ID—while separately making it more deferential to a conversational user. Which sign dominates is bias-specific, so “instruction tuning increases bias” should not be assumed as a general rule.

**A note on mechanism.** A natural conjecture is that instruction tuning improves how reliably the model maps a rubric label to a score, reducing format-driven noise. We deliberately do *not* present internal (e.g. attention-based) evidence for any mechanism: our data do not support such a claim, and we leave mechanistic analysis to future work with instrumented model runs.

## 6 Limitations

1. **Aggregate data.** Only per-variant mean scores were retained, so all inference is at the family level ( $n = 7$ ). Item-level metrics (flip rate, discrimination, domain effects) cannot be computed and are not reported.
2. **Small  $n$  and low power.** With seven families the exact Wilcoxon test cannot fall below  $p = 0.047$ , and a multiplicity correction across three probes would render it non-significant. We rely on effect sizes, bootstrap CIs, and directional consistency, and flag the rubric-order result as inconclusive.
3. **Scale.** All models are  $\leq 7B$ . Larger models—especially strong RLHF chat models—may behave differently; our data cannot speak to them, and we make no claim that the reduction persists at scale.
4. **Single recipe per family, English only, greedy decoding.** We cannot separate SFT from RLHF contributions, generalise beyond English, or estimate within-model variance (temperature 0 makes repeats deterministic).

## 7 Conclusion

Across seven small open-weight families, instruction tuning *reduces* the scoring biases of Li et al. (2026), robustly so for score-ID and reference-answer bias. This is directional evidence that these biases are shaped during instruction tuning rather than fixed at pre-training, and that, for small judges, tuning improves robustness rather than harming it. The claim is modest by construction: it rests on seven families of aggregate data. We hope the zero-cost, fully reproducible design lowers the barrier to the larger, item-level, larger-scale replication needed to settle the question.

## Reproducibility

Every statistic and figure in this paper is produced by `repro/analyze.py` and `repro/make_figures.py` from the single data file `t4fam_results.json` (seed 42). No synthetic or simulated data is used.

## Ethics Statement

No human subjects were involved. All models are publicly available open-weight checkpoints used under their respective licenses. The experiment runs at zero monetary cost on a free GPU tier.

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